

Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels. report findings must be backed by solid proof from within the data

****EXECUTIVE SUMMARY**** Skincare generates the most revenue (\$241,628) but its defect rate (2.34 %) is only marginally higher than haircare (2.48 %). Carrier C delivers the strongest revenue-to-shipping-cost ratio (1,138.6 \$ per shipping dollar), while 36 SKUs failed inspection, with several high-defect outliers (e.g., SKU42 at 4.94 %). Stockout exposure is acute for a handful of items, notably SKU34 (risk score 13) where a single unit on hand must cover a 26-day lead time.

****KEY FINDINGS**** - ****Product type defect hierarchy:**** Haircare = 2.483 % (highest), Skincare = 2.335 %, Cosmetics = 1.919 % (lowest). - ****Carrier efficiency:**** Carrier C yields the best revenue-per-shipping-dollar at **1,138.6 \$**, followed by Carrier B (**1,055.7 \$**) and Carrier A (**917 \$**). Carrier B also posts the lowest average defect rate (1.76 %). - ****Failed inspection SKUs:**** All 36 listed SKUs failed inspection; the top three defect rates are SKU42 (4.94 %), SKU65 (4.91 %), and SKU50 (4.75 %). These SKUs span skincare and cosmetics and are shipped primarily via Carrier C and Carrier A. - ****Stockout risk:**** The highest risk score is SKU34 (score 13) with 1 unit in stock and a 26-day lead time. Other critical items include SKU68 (score 8, zero stock) and SKU24 (score 5.2, 4 units, 26-day lead time). - ****Margin stability:**** Gross-margin ratios are uniformly high across product lines (cosmetics 0.993, haircare 0.991, skincare 0.992), indicating pricing power despite defect and logistics variances.

****RISKS AND RECOMMENDATIONS**** The convergence of elevated defect rates in haircare, high-defect SKUs (especially those on Carrier C), and prolonged lead times for low-stock items creates a material risk to service levels and brand reputation. Immediate actions should include: (1) tightening quality controls on the 36 failed inspection SKUs and prioritizing corrective actions for the top-defect outliers; (2) renegotiating carrier contracts to shift more volume to Carrier B, which combines the best defect performance with strong revenue efficiency; and (3) implementing safety-stock buffers for the top-risk SKUs (e.g., SKU34, SKU68) to mitigate stockout exposure while reviewing lead-time reductions with manufacturers.